FED. SUP CLASS
6630



PART NUMBER	NOMINAL API GRAVITY RANGE (DEGREES)	THERMOMETER RANGE (°F)
MS36539-1	-1 to 11	20 to 130
MS36539-2	9 to 21	20 to 130
MS36539-3	19 to 31	20 to 130
MS36539-4	29 to 41	20 to 130
MS36539-5	39 to 51	20 to 130
MS36539-6	49 to 61	0 to 110
MS36539-7	59 to 71	0 to 110
MS36539-8	69 to 81	0 to 110

MATERIAL: (A) Hydrometers shall be of borosilicate glass in accordance with ASTM E 438, Type I, Class A.

Ballast shall be of mercury, lead shot or fused metal alloy.

Scales shall be printed on No. 1 white sulfite or white ledger paper.

Graduation lines and inscriptions shall be made with permanent black marking material, such as India ink.

DESIGN: Hydrometers shall be of the constant-mass, variable displacement type, and shall have a stem, a body, and a thermometer in the body.

(A) Hydrometers shall have an overall length of 190 ± 5 mm, and the bulb shall have a diameter of 14 ± 2 mm.

(A) The stem shall have a securely fastened scale graduated in American Petroleum Institute (API) gravity degrees in 0.2° increments with major graduations and numbers at every 1° API.

Ballast shall be securely fastened in the lower portion of the body.

Standard temperature for calibration shall be at 60°F.

Error in the API gravity scale shall not exceed 0.2 degrees.

Thermometer shall be fastened in the body with the ballast.

Thermometer shall be graduated in 2°F increments with major graduations at 10°F and with every 20°F increments numbered.

Fahrenheit temperature scale shall not have an error greater than 1°F.

PURPOSE: For determining the API gravity of petroleum products and other liquids of similar surface tension (33 dynes per centimeter or less).

NOTES:
1. Hydrometers applicable to ASTM E 100 are considered standard for military use under this document.
2. Items differing from this Military Standard by minor variations in design or dimensions are considered standard under this document, provided such variations do not affect performance and are not in conflict with the material requirements as specified herein.

APPROVED 7 OCT 1968 REVISED (A) 10 JUN 1986

USER ACTIVITIES: ARMY - AV, AR, ER, ME
NAVY - CG, OS, YD

REVIEWER ACTIVITIES: ARMY - MD
NAVY - MS
AIR FORCE - 03

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P.A. EA	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
Other Cust MS 99		HYDROMETER, GRADUATED SCALE, API GRAVITY (WITH THERMOMETER)	MS36539
Procurement Specifications MIL-H-36343	SUPERSEDES:		SHEET 1 OF 2

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FED. SUP CLASS
6630

3. For design feature purposes, this standard takes precedence over procurement document referenced herein.

4. Referenced documents shall be of the issue in effect on date of invitations for bid.

Ⓐ 5. FSCMs: 11876, 53629, and 64484.

USER ACTIVITIES:

REVIEWER ACTIVITIES:

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